```
bhargav@Bhargava:/mnt/c/Users/bharg/Downloads/College/CN-NP-2-2$ sudo ip netns add host1
sudo ip netns add router
sudo ip netns add host2

sudo ip link add veth-h1 type veth peer name veth-r1
sudo ip link add veth-r2 type veth peer name veth-h2

sudo ip link set veth-h1 netns host1
sudo ip link set veth-r1 netns router
sudo ip link set veth-r2 netns router
sudo ip link set veth-h2 netns host2
```

```
bhargav@Bhargava:/mnt/c/Users/bharg/Downloads/College/CN-NP-2-2$ # host1
sudo ip netns exec host1 ip addr add 10.0.1.1/30 dev veth-h1
sudo ip netns exec host1 ip link set veth-h1 up
sudo ip netns exec host1 ip link set lo up

# router
sudo ip netns exec router ip addr add 10.0.1.2/30 dev veth-r1
sudo ip netns exec router ip addr add 10.0.2.1/30 dev veth-r2
sudo ip netns exec router ip link set veth-r1 up
sudo ip netns exec router ip link set veth-r2 up
sudo ip netns exec router ip link set lo up

# host2
sudo ip netns exec host2 ip addr add 10.0.2.2/30 dev veth-h2
sudo ip netns exec host2 ip link set veth-h2 up
sudo ip netns exec host2 ip link set lo up
```

```
bhargav@Bhargava:/mnt/c/Users/bharg/Downloads/College/CN-NP-2-2$ sudo ip netns exec router sysctl -w net.ipv4.ip_forward=1
sudo ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2
sudo ip netns exec host2 ip route add 10.0.1.0/30 via 10.0.2.1
net.ipv4.ip_forward = 1
bhargav@Bhargava:/mnt/c/Users/bharg/Downloads/College/CN-NP-2-2$ sudo ip netns exec host1 ping -c 3 10.0.2.2
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
64 bytes from 10.0.2.2: icmp_seq=1 ttl=63 time=0.222 ms
64 bytes from 10.0.2.2: icmp_seq=2 ttl=63 time=0.066 ms
64 bytes from 10.0.2.2: icmp_seq=3 ttl=63 time=0.066 ms

--- 10.0.2.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2056ms
rtt min/avg/max/mdev = 0.066/0.118/0.222/0.073 ms
bhargav@Bhargava:/mnt/c/Users/bharg/Downloads/College/CN-NP-2-2$
```

```
bhargav@Bhargava:/mnt/c/Users/bharg/Downloads/College/CN-NP-2-2$ sudo ip netns exec router tc qdisc add dev veth-r2 root netem delay 20ms loss 1%
sudo ip netns exec router tc qdisc show dev veth-r2
qdisc netem 8001: root refcnt 17 limit 1000 delay 20ms loss 1% seed 2291019968257398711
bhargav@Bhargava:/mnt/c/Users/bharg/Downloads/College/CN-NP-2-2$
```

```
bhargav@Bhargava:/mnt/c/Users/bharg/Downloads/College/CN-NP-2-2$ sudo ip netns exec host1 nc 10.0.2.2 1234
[sudo: authenticate] Password:
Hello from host1!
This is Bharagav
Three Point Capture Lab
Message going through router
```

wsl
wsl
wsl
wsl
wsl

```
bhargav@Bhargava:/mnt/c/Users/bharg/Downloads/College/CN-NP-2-2$ sudo ip netns exec host2 nc -l 1234
[sudo: authenticate] Password:
Hello from host1!
This is Bharagav
Three Point Capture Lab
Message going through router
```

```
bhargav@Bhargava:/mnt/c/Users/bharg/Downloads/College/CN-NP-2-2$ sudo ip netns exec host1 tshark -i veth-h1 -w /tmp/cap1.pcap
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-h1'
24 ^C
bhargav@Bhargava:/mnt/c/Users/bharg/Downloads/College/CN-NP-2-2$
```

```
bhargav@Bhargava:/mnt/c/Users/bharg/Downloads/College/CN-NP-2-2$ sudo ip netns exec router tshark -i veth-r1 -w /tmp/cap2.pcap
[sudo: authenticate] Password:
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-r1'
23 ^C
bhargav@Bhargava:/mnt/c/Users/bharg/Downloads/College/CN-NP-2-2$
```

```
bhargav@Bhargava:/mnt/c/Users/bharg/Downloads/College/CN-NP-2-2$ sudo ip netns exec host2 tshark -i veth-h2 -w /tmp/cap3.pcap
[sudo: authenticate] Password:
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-h2'
17 ^C
bhargav@Bhargava:/mnt/c/Users/bharg/Downloads/College/CN-NP-2-2$
```

```
bhargav@Bhargava:/mnt/c/Users/bharg/Downloads/College/CN-NP-2-2$ sudo chmod 644 /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap
ls -lh /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap
-rw-r--r-- 1 root root 2.7K Nov 10 10:46 /tmp/cap1.pcap
-rw-r--r-- 1 root root 2.6K Nov 10 10:46 /tmp/cap2.pcap
-rw-r--r-- 1 root root 2.1K Nov 10 10:46 /tmp/cap3.pcap
bhargav@Bhargava:/mnt/c/Users/bharg/Downloads/College/CN-NP-2-2$
```

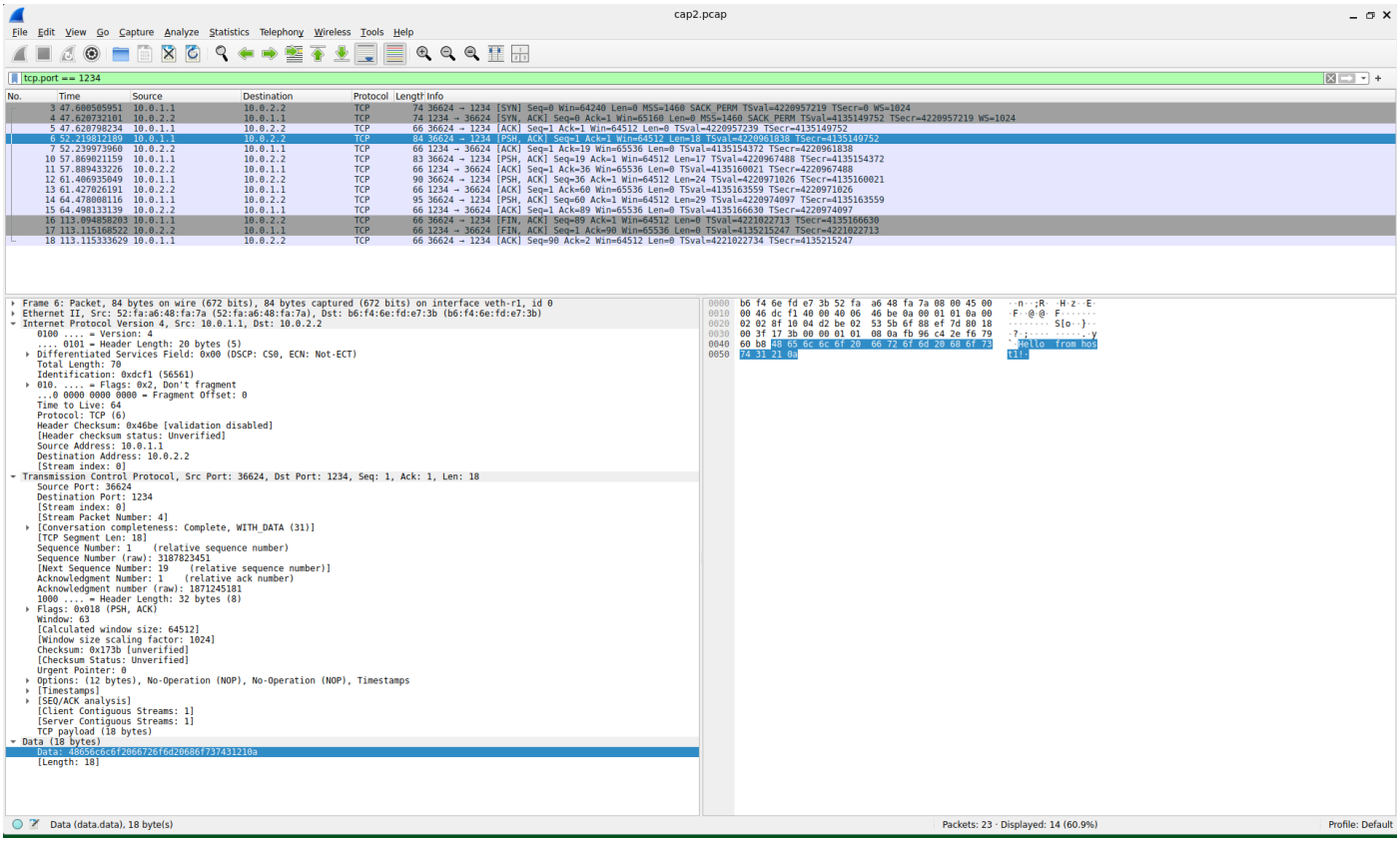
```
bhargav@Bhargava:/mnt/c/Users/bharg/Downloads/College/CN-NP-2-2$ tshark -r /tmp/cap1.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c

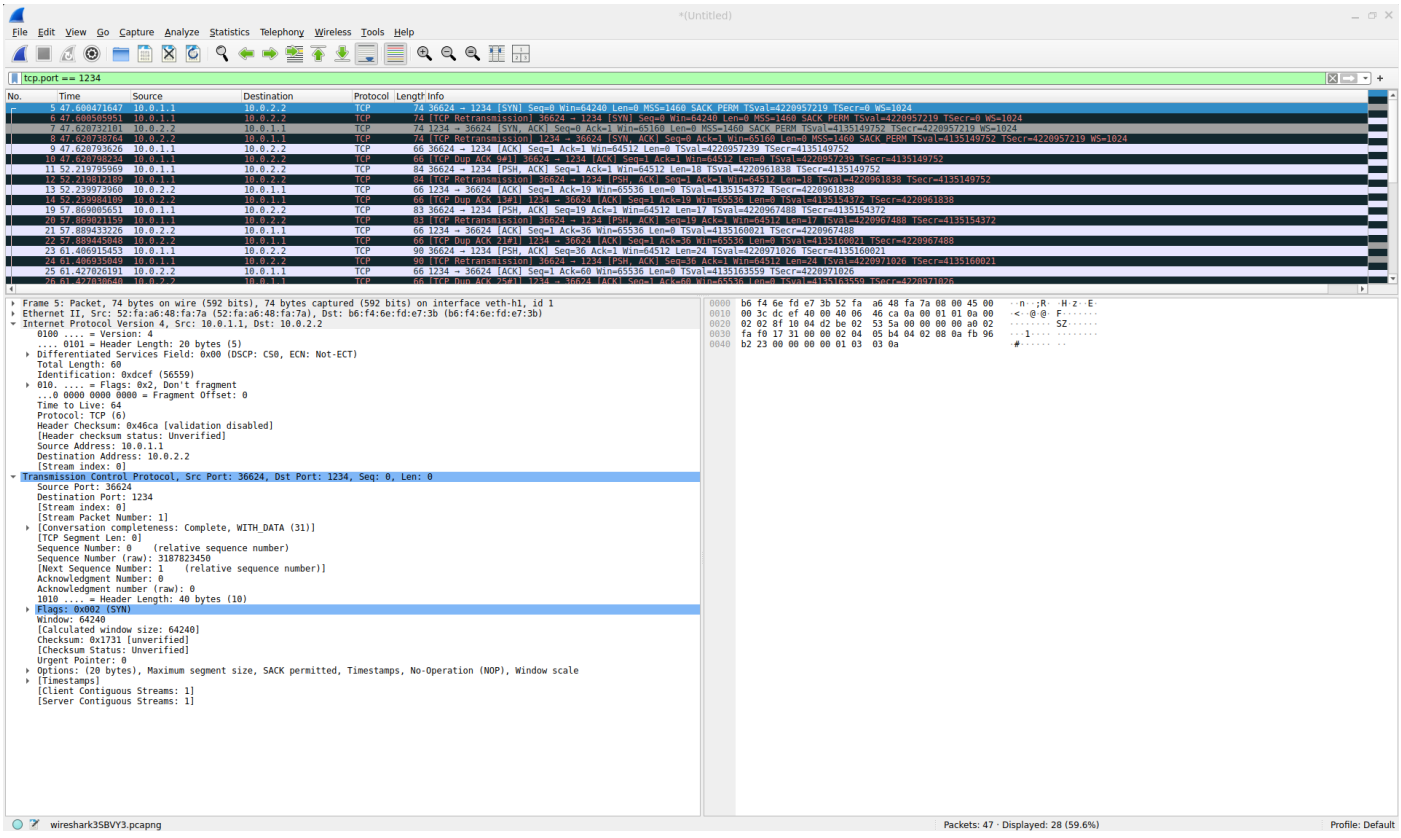
tshark -r /tmp/cap2.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
tshark -r /tmp/cap3.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
      6 63
      8 64
      6 63
      8 64
      8 63
      6 64
bhargav@Bhargava:/mnt/c/Users/bharg/Downloads/College/CN-NP-2-2$
```

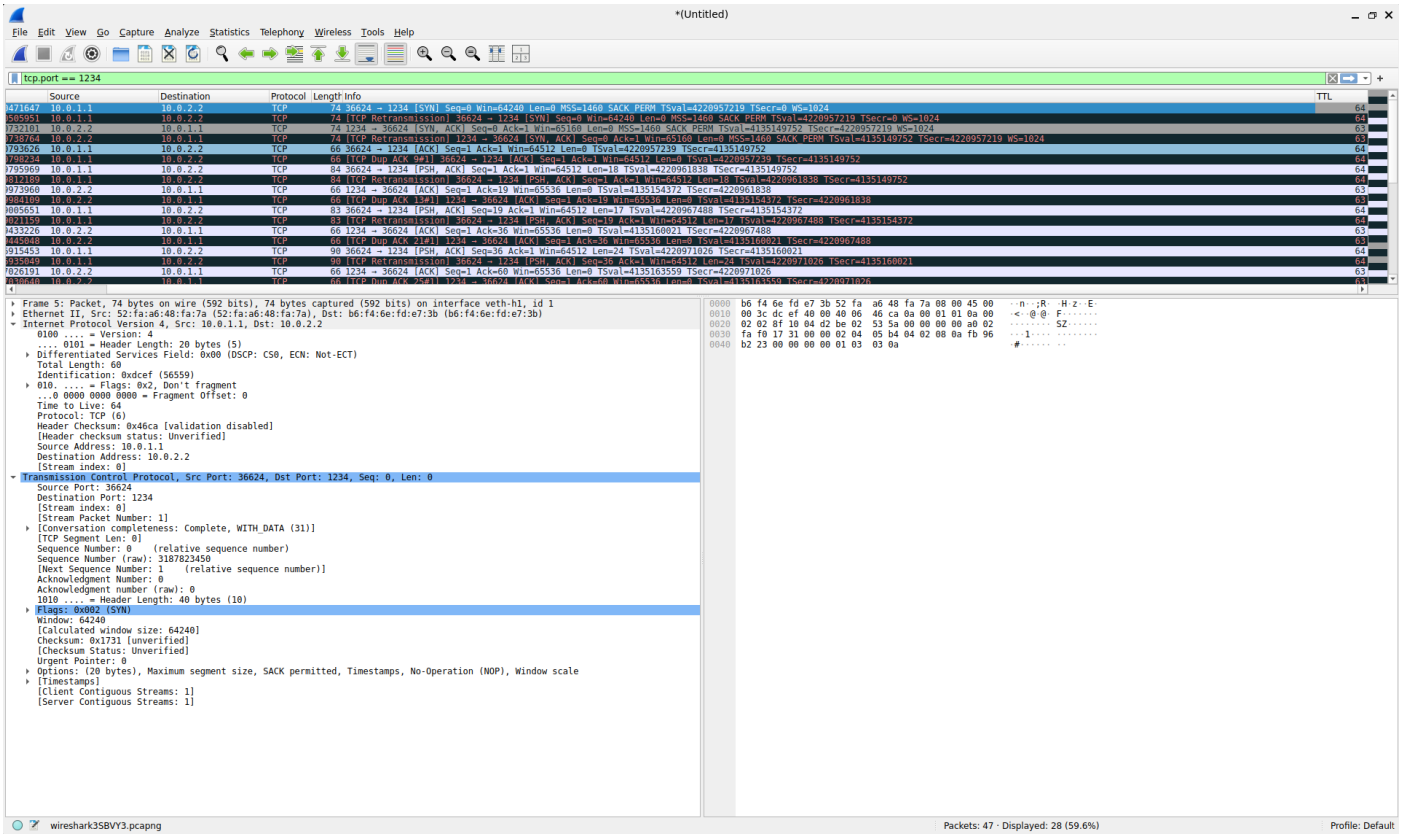
```
bhargav@Bhargava:/mnt/c/Users/bharg/Downloads/College/CN-NP-2-2$ tshark -r /tmp/cap2.pcap -Y "tcp.port==1234" -T fields -e tcp.payload 2>/dev/null
 | xxd -r -p
Hello from host1!
This is Bharagav
Three Point Capture Lab
Message going through router
bhargav@Bhargava:/mnt/c/Users/bharg/Downloads/College/CN-NP-2-2$
```

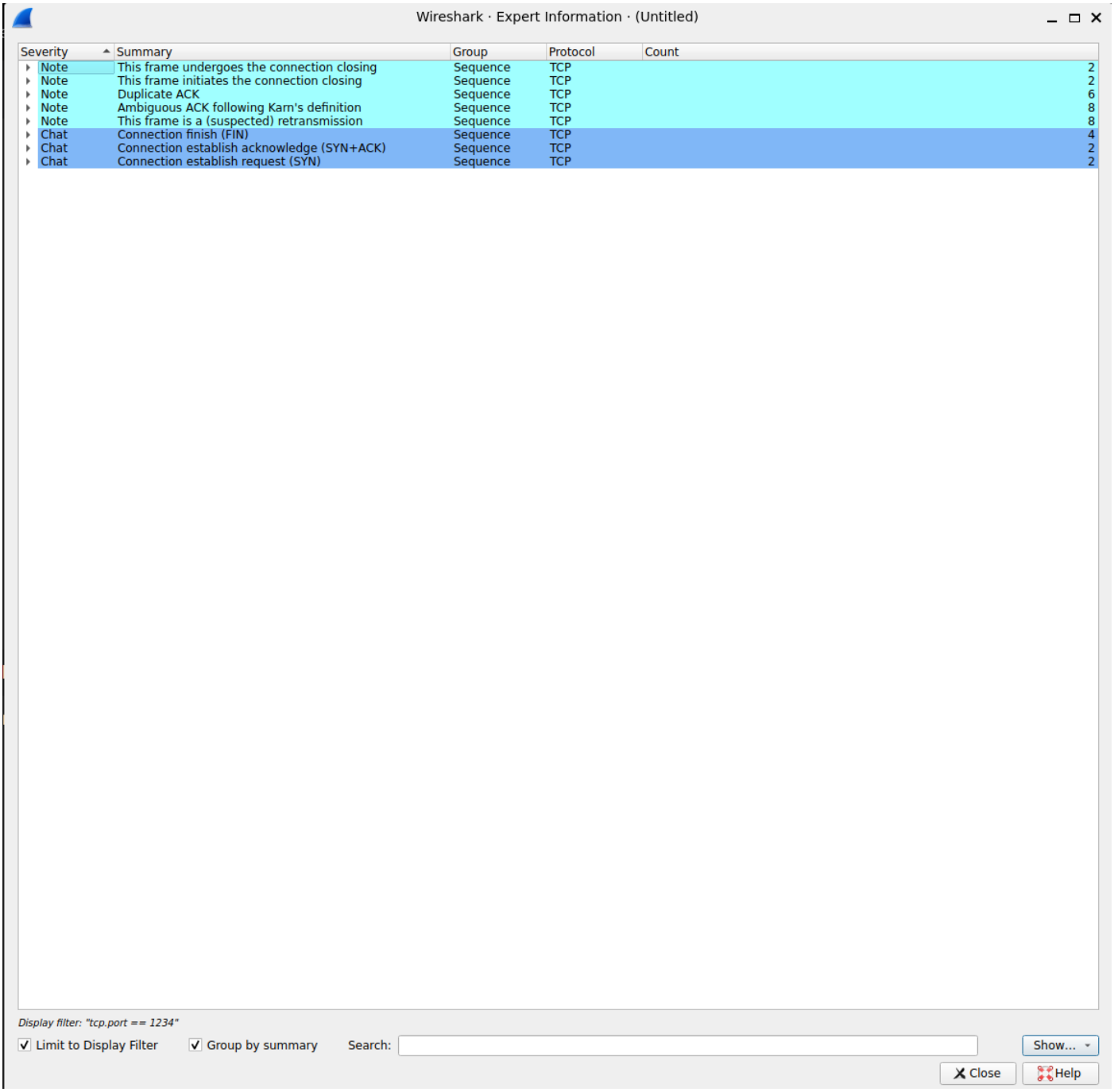
```
bhargav@Bhargava:/mnt/c/Users/bharg/Downloads/College/CN-NP-2-2$ tshark -r /tmp/cap1.pcap -Y "tcp.analysis.retransmission" 2>/dev/null | wc -l
tshark -r /tmp/cap2.pcap -Y "tcp.analysis.retransmission" 2>/dev/null | wc -l
tshark -r /tmp/cap3.pcap -Y "tcp.analysis.retransmission" 2>/dev/null | wc -l
sudo ip netns exec host1 mtr --report 10.0.2.2
0
0
0
Start: 2026-11-10T10:48:56+0000

HOST: Bhargava                    Loss%   Snt   Last   Avg  Best  Wrst StDev
  1.|-- 10.0.1.2                   0.0%    10    0.1   0.1   0.1   0.5   0.1
  2.|-- 10.0.2.2                   0.0%    10   20.3  20.4  20.2  20.8   0.2
bhargav@Bhargava:/mnt/c/Users/bharg/Downloads/College/CN-NP-2-2$
```




*(Untitled)
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp.port == 1234
No. Time Source Destination Protocol Length Info
5 47.600471647 10.0.1.1 10.0.2.2 TCP 74 36624 → 1234 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=4220957219 TSecr=0 WS=1024
6 47.600505951 10.0.1.1 10.0.2.2 TCP 74 [TCP Retransmission] 36624 → 1234 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=4220957219 TSecr=0 WS=1024
7 47.620732101 10.0.2.2 10.0.1.1 TCP 74 1234 → 36624 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM TSval=4135149752 TSecr=4220957219 WS=1024
8 47.620738764 10.0.2.2 10.0.1.1 TCP 74 [TCP Retransmission] 1234 → 36624 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM TSval=4135149752 TSecr=4220957219 WS=1024
9 47.620793626 10.0.1.1 10.0.2.2 TCP 66 36624 → 1234 [ACK] Seq=1 Ack=1 Win=64512 Len=0 TSval=4220957239 TSecr=4135149752
10 47.620796214 10.0.1.1 10.0.2.2 TCP 66 [TCP Dup ACK 9#1] 36624 → 1234 [ACK] Seq=1 Ack=1 Win=64512 Len=0 TSval=4220957239 TSecr=4135149752
11 52.219795969 10.0.1.1 10.0.2.2 TCP 84 36624 → 1234 [PSH, ACK] Seq=1 Ack=1 Win=64512 Len=18 TSval=4220961838 TSecr=4135149752
12 52.219812189 10.0.1.1 10.0.2.2 TCP 84 [TCP Retransmission] 36624 → 1234 [PSH, ACK] Seq=1 Ack=1 Win=64512 Len=18 TSval=4220961838 TSecr=4135149752
13 52.239973960 10.0.2.2 10.0.1.1 TCP 66 1234 → 36624 [ACK] Seq=1 Ack=19 Win=65536 Len=0 TSval=4135154372 TSecr=4220961838
14 52.239984109 10.0.2.2 10.0.1.1 TCP 66 [TCP Dup ACK 13#1] 1234 → 36624 [ACK] Seq=1 Ack=19 Win=65536 Len=0 TSval=4135154372 TSecr=4220961838
19 57.869005651 10.0.1.1 10.0.2.2 TCP 83 36624 → 1234 [PSH, ACK] Seq=19 Ack=1 Win=64512 Len=17 TSval=4220967488 TSecr=4135154372
20 57.869021159 10.0.1.1 10.0.2.2 TCP 83 [TCP Retransmission] 36624 → 1234 [PSH, ACK] Seq=19 Ack=1 Win=64512 Len=17 TSval=4220967488 TSecr=4135154372
21 57.889433226 10.0.2.2 10.0.1.1 TCP 66 1234 → 36624 [ACK] Seq=1 Ack=36 Win=65536 Len=0 TSval=4135160021 TSecr=4220967488
22 57.889441049 10.0.2.2 10.0.1.1 TCP 66 [TCP Dup ACK 21#1] 1234 → 36624 [ACK] Seq=1 Ack=36 Win=65536 Len=0 TSval=4135160021 TSecr=4220967488
23 61.406915453 10.0.1.1 10.0.2.2 TCP 90 36624 → 1234 [PSH, ACK] Seq=36 Ack=1 Win=64512 Len=24 TSval=4220971026 TSecr=4135160021
24 61.406935049 10.0.1.1 10.0.2.2 TCP 90 [TCP Retransmission] 36624 → 1234 [PSH, ACK] Seq=36 Ack=1 Win=64512 Len=24 TSval=4220971026 TSecr=4135160021
25 61.427026191 10.0.2.2 10.0.1.1 TCP 66 1234 → 36624 [ACK] Seq=1 Ack=60 Win=65536 Len=0 TSval=4135163559 TSecr=4220971026
26 61.427030648 10.0.2.2 10.0.1.1 TCP 66 [TCP Dup ACK 25#1] 1234 → 36624 [ACK] Seq=1 Ack=60 Win=65536 Len=0 TSval=4135163559 TSecr=4220971026
Frame 5: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface veth-h1, id 1
Ethernet II, Src: 52:fa:a6:48:fa:7a (52:fa:a6:48:fa:7a), Dst: b6:f4:6e:fd:e7:3b (b6:f4:6e:fd:e7:3b)
Internet Protocol Version 4, Src: 10.0.1.1, Dst: 10.0.2.2
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 60
Identification: 0xdcef (56559)
010. .... = Flags: 0x2, Don't fragment
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 64
Protocol: TCP (6)
Header Checksum: 0x46ca [validation disabled]
[Header checksum status: Unverified]
Source Address: 10.0.1.1
Destination Address: 10.0.2.2
[Stream index: 0]
Transmission Control Protocol, Src Port: 36624, Dst Port: 1234, Seq: 0, Len: 0
Source Port: 36624
Destination Port: 1234
[Stream index: 0]
[Stream Packet Number: 1]
[Conversation completeness: Complete, WITH_DATA (31)]
[TCP Segment Len: 0]
Sequence Number: 0 (relative sequence number)
Sequence Number (raw): 3187823450
[Next Sequence Number: 1 (relative sequence number)]
Acknowledgment Number: 0
Acknowledgment number (raw): 0
1010 .... = Header Length: 40 bytes (10)
Flags: 0x002 (SYN)
Window: 64240
[Calculated window size: 64240]
Checksum: 0x1731 [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
Options: (20 bytes), Maximum segment size, SACK permitted, Timestamps, No-Operation (NOP), Window scale
[Timestamps]
[Client Contiguous Streams: 1]
[Server Contiguous Streams: 1]
0000 b6 f4 6e fd e7 3b 52 fa a6 48 fa 7a 08 00 45 00 ··n··;R· ·H·z··E·
0010 00 3c dc ef 40 00 40 06 46 ca 0a 00 01 01 0a 00 ·<··@·@· F·····
0020 02 02 8f 10 04 d2 be 02 53 5a 00 00 00 00 a0 02 ········ SZ·····
0030 fa f0 17 31 00 00 02 04 05 b4 04 02 08 0a fb 96 ···1···· ········
0040 b2 23 00 00 00 00 01 03 03 0a ·#·····
wireshark3SBVY3.pcapng
Packets: 47 · Displayed: 28 (59.6%)
Profile: Default


*(Untitled)
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp.port == 1234
Source Destination Protocol Length Info TTL
Frame 5: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface veth-h1, id 1
Ethernet II, Src: 52:fa:a6:48:fa:7a (52:fa:a6:48:fa:7a), Dst: b6:f4:6e:fd:e7:3b (b6:f4:6e:fd:e7:3b)
Internet Protocol Version 4, Src: 10.0.1.1, Dst: 10.0.2.2
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 60
Identification: 0xdcef (56559)
010. .... = Flags: 0x2, Don't fragment
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 64
Protocol: TCP (6)
Header Checksum: 0x46ca [validation disabled]
[Header checksum status: Unverified]
Source Address: 10.0.1.1
Destination Address: 10.0.2.2
[Stream index: 0]
Transmission Control Protocol, Src Port: 36624, Dst Port: 1234, Seq: 0, Len: 0
Source Port: 36624
Destination Port: 1234
[Stream index: 0]
[Stream Packet Number: 1]
[Conversation completeness: Complete, WITH_DATA (31)]
[TCP Segment Len: 0]
Sequence Number: 0 (relative sequence number)
Sequence Number (raw): 3187823450
[Next Sequence Number: 1 (relative sequence number)]
Acknowledgment Number: 0
Acknowledgment number (raw): 0
1010 .... = Header Length: 40 bytes (10)
Flags: 0x002 (SYN)
Window: 64240
[Calculated window size: 64240]
Checksum: 0x1731 [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
Options: (20 bytes), Maximum segment size, SACK permitted, Timestamps, No-Operation (NOP), Window scale
[Timestamps]
[Client Contiguous Streams: 1]
[Server Contiguous Streams: 1]
0000 b6 f4 6e fd e7 3b 52 fa a6 48 fa 7a 08 00 45 00 ··n··;R· ·H·z··E·
0010 00 3c dc ef 40 00 40 06 46 ca 0a 00 01 01 0a 00 ·<··@·@· F·····
0020 02 02 8f 10 04 d2 be 02 53 5a 00 00 00 00 a0 02 ········ SZ·····
0030 fa f0 17 31 00 00 02 04 05 b4 04 02 08 0a fb 96 ···1···· ········
0040 b2 23 00 00 00 00 01 03 03 0a ·#·····
wireshark3SBVY3.pcapng
Packets: 47 · Displayed: 28 (59.6%)
Profile: Default




```
bhargav@Bhargava:/mnt/c/Users/bharg/Downloads/College/CN-NP-2-2$ sudo ip netns del host1
sudo ip netns del router
sudo ip netns del host2
sudo ip netns list
[1]+  Done                    wireshark /tmp/cap2.pcap
```